

Describing Meaning – Week 2: Sentences, Connectives and Propositional Logic

1. Roadmap and framing

Today

- **Last time:** truth conditions, entailment, tautologies/contradictions.
- **This time:**
 1. Complex sentences
 2. Symbolisation
 3. Truth-functional connectives
 4. Truth tables
 5. Cross-linguistic variation in connective systems

Framing

- **Phenomenon:** complex sentences are built from simpler ones.
 - *not, and, or, if... then*
 - and they support stable entailment patterns.
- **Formal tools:** propositional logic, symbolisation, truth tables.
- **Cross-linguistic comparison:**
 - Latin disjunction markers (*aut, vel*, enclitic *-ve*, etc.)
 - how languages realise (and partition) connective inventories.

2. Entailment and complex sentences

Entailments with predicates

Some **quantificational** entailments follow from set-theoretic structure.

- All cats stalk. $Cats \subseteq Stalks$
- Tilly is a cat. $t \in Cats$
- Therefore, Tilly stalks. $t \in Stalks$

We don't need world knowledge here: if the first two are true, the third must be true.

Formally: A **entails** B iff **there is no way for A to be true and B false at the same time.**

Complex sentences

Another place where we see stable patterns are **complex sentences**, built from simpler ones.

- “**It is not the case that** it is raining.”
- “It is raining **and** I have an umbrella.”
- “**Either** he'll drink Guinness **or** he'll drink a lager.”
- “**If** he drinks too much, **then** he won't come to class tomorrow.”

Key idea: the truth of the complex sentence is a **function of the truth** of its parts.

Sentential entailments

- From a conjunction you can infer either conjunct (**Conjunctive Simplification**).
 - It is raining and I have an umbrella.
 - $\therefore$ It is raining.
- From a conditional and its antecedent you can infer its consequent (**Modus Ponens**).

- If he drank too much last night, then he won't be in class.
- He drank too much last night.
- $\therefore$ He won't be in class.
- From a disjunction and the negation of one disjunct, infer the other (**Modus Tolendo Ponens** / Disjunctive Syllogism).
 - He'll drink Guinness or he'll drink a lager.
 - He's not drinking a lager.
 - $\therefore$ He's drinking Guinness.
- From a double negation infer the non-negated sentence (**Double Negation Elimination**).
 - He doesn't **not** know what he's doing.
 - $\therefore$ He knows what he's doing.

3. Symbolisation

Why symbolise?

We capture these entailments by **translating** English into a formal language where validity follows from the system.

The relevant logic is called **sentential** / **propositional** / **truth-functional** logic.

Goal: show how truth of complex sentences is **determined** by truth of parts (truth tables).

Atomic propositions

- Identify **atomic sentences** (can't be further analyzed here) and map them to **propositional variables** $p, q, r, \dots$
- "It is raining" = p

- "It is raining and I have an umbrella"
 - "It is raining" = p
 - "I have an umbrella" = q
- Next: symbolise the **connectives** that build complex sentences from atoms.

4. Truth-functional connectives and basic truth tables

The connectives

Assuming p and q are propositions:

- **Conjunction:** $p \wedge q$ (p and q)
- **Disjunction:** $p \vee q$ (p or q)
- **Material implication:** $p \rightarrow q$ (*if p , then q*)
- **Negation:** $\neg p$ (*not p*)

Negation

- (1) "It is not the case that it is raining."

- $\neg p$ is true iff p is false.

p	$\neg p$
1	0
0	1

Conjunction

(2) “It is raining **and** I have an umbrella.”

- The conjunction is true iff both conjuncts are true ($p \wedge q$).

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0

Disjunction (inclusive vs. exclusive)

(3) “**Either** he’ll drink Guinness **or** he’ll drink a lager.”

- Inclusive *or*: $p \vee q$ is true if at least one disjunct is true.
- Exclusive *or* (XOR): true iff exactly one disjunct is true.

Inclusive $p \vee q$		
p	q	$p \vee q$
1	1	1
1	0	1
0	1	1
0	0	0

Exclusive $p \text{ XOR } q$		
p	q	$p \text{ XOR } q$
1	1	0
1	0	1
0	1	1
0	0	0

The material conditional

- The connective $p \rightarrow q$ is read as “*if p , then q* ”.
- In propositional logic this is interpreted as the **material conditional**.
- It is defined as false only in the case where p is true and q is false.

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

The biconditional

- The connective $p \leftrightarrow q$ is read as “ *p iff q* ” / “ *p if and only if q* ”.
- It is defined as true exactly when p and q have the **same** truth value.

p	q	$p \leftrightarrow q$
1	1	1
1	0	0
0	1	0
0	0	1

5. Cross-linguistic comparison: connective systems

How connective systems are realised

Languages vary in how they realise “connectives” in form. Some common strategies:

- free words/particles (like English *and*, *or*)
- correlatives (*either... or*, Latin *aut... aut*)
- clitics / enclitics (Latin *-ve*)
- prosody / discourse particles (often relevant for conditionals and alternatives)

Even when form differs, the **truth-conditional patterns** are often stable enough to model with truth tables.

Latin *aut* vs. *vel*: a tempting story

A familiar **textbook line** (especially in logic pedagogy) is:

- *vel* = “weak” / inclusive *or*
- *aut* = “strong” / exclusive *or*

This gives a neat mapping from Latin particles to truth tables. But as a claim about **lexical semantics**, it is probably too neat.

A more likely view: exclusivity as pragmatic strengthening

A better default hypothesis is:

- the basic truth-conditional core of disjunction is **inclusive**
- “exclusive” readings often arise by **pragmatic strengthening**.

aut commonly appears in **contrastive/listing** environments where alternatives are treated as competing or disjoint. In those environments, exclusivity can be a consequence of discourse structure (salient alternatives, expectations), rather than an XOR meaning in the lexicon.

Also: Latin has **more than two** disjunction markers (*aut*, *vel*, enclitic *-ve*, *sive/seu*, ...).

6. Truth-functional completeness and typology

Truth-functional completeness

A set of connectives is **truth-functionally complete** if you can express any truth table (any truth function) using just those connectives.

- Example: $\{\neg, \wedge, \vee\}$ is complete.
- As it turns out so is $\{\neg, \wedge\}$ and $\{\neg, \vee\}$.
- Surprisingly two single connectives are complete all by themselves:
 - **NAND** (Sheffer stroke): $p \uparrow q \equiv \neg(p \wedge q)$

– **NOR**: $p \downarrow q \equiv \neg(p \vee q)$

A typological observation: no lexicalised NAND

If NAND is so expressive, you might expect languages to have a basic word meaning NAND. But (as far as we know) **no natural language lexicalises NAND as a simple connective**. Instead, languages express it compositionally:

- *not* (*p and q*)
- *not both p and q*
- *it's not the case that p and q*

So there may be constraint or universal stopping NAND from being lexicalised.

7. Complex truth tables

Exercises: complex truth tables

Draw complete truth tables for each of the following formulas:

- $\neg(p \wedge q)$
- $(\neg p \wedge q)$
- $(p \wedge q) \vee r$
- $p \wedge (q \vee r)$
- $p \rightarrow (q \wedge r)$
- $(p \rightarrow q) \wedge r$
- $(p \rightarrow r) \wedge (q \rightarrow r)$

8. Tautologies and contradictions

Definitions

- **Tautologies:** formulas that are true no matter what truth values we assign to the atomic propositions.
- Examples: $p \rightarrow p$, $p \vee \neg p$, $\neg(p \wedge \neg p)$.
- **Contradictions:** formulas that are false no matter what truth values we assign.
- Examples: $p \wedge \neg p$, $p \rightarrow \neg(p \vee q)$.

Exercises

- Verify with truth tables that $p \vee \neg p$ is a tautology.
- Show that $p \wedge \neg p$ is a contradiction.
- Decide whether $(p \rightarrow q) \vee (q \rightarrow p)$ is a tautology, contradiction, or contingent.

9. Entailment, validity, and testing with truth tables

Entailment and validity

- **Entailment:** $\Gamma \models \varphi$ iff there is no row where all formulas in Γ are true and φ is false.
- Special case (one premise): $A \models B$ iff there is no row where $A = 1$ and $B = 0$.
- **Validity:** an argument is valid iff its premises entail its conclusion (i.e. the same truth-table test: no counterexample row).
- Multiple premises can be combined by conjoining them.

- Classic valid inference patterns:

- Modus Ponens: $p \rightarrow q, p \therefore q$
- Modus Tolendo Ponens (Disjunctive Syllogism): $p \vee q, \neg q \therefore p$

Exercises

- Test whether $p \wedge q$ entails p .
- Test whether p entails $p \vee q$.
- Use a truth table to show that Modus Ponens is valid.
- Use a truth table to show that Modus Tolendo Ponens is valid.
- Does $p \rightarrow q$ entail $\neg p \vee q$? Prove using a truth table.

10. Proving equivalences

Methods

To show $\varphi \equiv \psi$, we can:

- Show the truth tables are identical.
- Show mutual entailment.
- Show $(\varphi \leftrightarrow \psi)$ is a tautology.

Exercises

- Prove $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ by one of the three methods.
- Show that $p \leftrightarrow q$ and $p \vee q$ are **not** equivalent (e.g. by giving a truth assignment where they differ, or by showing the truth tables differ).
- Show that $(p \vee q) \leftrightarrow (\neg p \rightarrow q)$ is a tautology.

11. De Morgan's Laws

Statement

Augustus De Morgan proved key equivalences involving negation, conjunction, and disjunction:

$$\neg(p \wedge q) \equiv (\neg p \vee \neg q) \quad \neg(p \vee q) \equiv (\neg p \wedge \neg q)$$

Natural language illustrations

- “I didn’t have both chips and a salad” $\Rightarrow$ “I didn’t have chips or I didn’t have a salad.”
- “I’m not French or Spanish” $\Rightarrow$ “I’m not French and I’m not Spanish.”

In words: *The negation of a conjunction is the disjunction of the negations; the negation of a disjunction is the conjunction of the negations.*

Exercises

- Prove $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$ using a truth table.
- Prove $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$ using a truth table.
- Give your own natural language example for each law and formalise it.

12. Practice: symbolisation and reading off truth

Practice: symbolisation & reading off truth

- Atomic propositions + connectives.
- “It is raining and I have an umbrella.”
 - p : It is raining q : I have an umbrella
 - $p \wedge q$
- Try these:
 - “It is raining and I don’t have an umbrella.”
 - “It isn’t raining and I don’t have an umbrella.”
 - “It is not the case that it is raining and I have an umbrella.”

More examples

- Translate: “If it rains, the match will be canceled.” $p \rightarrow q$
- Translate: “It is not both raining and cold.” $\neg(p \wedge q)$
- Translate: “Either Mary came or John left, but not both.” $(p \vee q) \wedge \neg(p \wedge q)$